GRADUATION PROJECT

Holiday Homes

Intelligent Rental Price Prediction System for Short-Term Tourist Properties

Prepared by: Wael Hlal

Specialization: Artificial Intelligence / Data Science

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About the Author

Name
Wael Hlal

Professional Experience
Over 10 years in legal and humanitarian work

Technical Certification
Certified AI Developer
Mystro GmbH — Germany

Specialization
Artificial Intelligence / Data Science

Technical Skills

- Flask
- Python
- LightGBM
- Pandas
- Scikit-learn
- Data Analysis
- Machine Learning
- Deep Learning

Languages

- Arabic — Native
- English — Fluent
- German — B2 Level

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Research Problem & Challenges

Underpricing

Leads to lost potential revenue and failure to achieve optimal returns from the property

Overpricing

Weakens competitiveness and negatively impacts occupancy rates and guest attraction

Manual Decision-Making

Relies on personal experience without leveraging the vast available data resources

Core Problem

The research problem lies in the absence of an integrated intelligent system capable of analyzing property characteristics and market conditions, then providing an expected price that supports managerial decisions within Holiday Homes companies. The project does not merely focus on building a mathematical prediction model — it seeks to deliver a practical framework that can be transformed into an operational tool helping companies make more accurate and consistent decisions across multiple cities and diverse markets.

Project Introduction & Objectives

Overview

This project aims to develop an intelligent system that serves short-term tourist property rental and management companies by predicting the expected rental price based on property characteristics and the surrounding market.

The project is built on real-world data from the Airbnb platform, focusing on analyzing pricing patterns across ten global cities. Its significance lies in shifting the pricing decision from personal judgment to a data-driven approach.

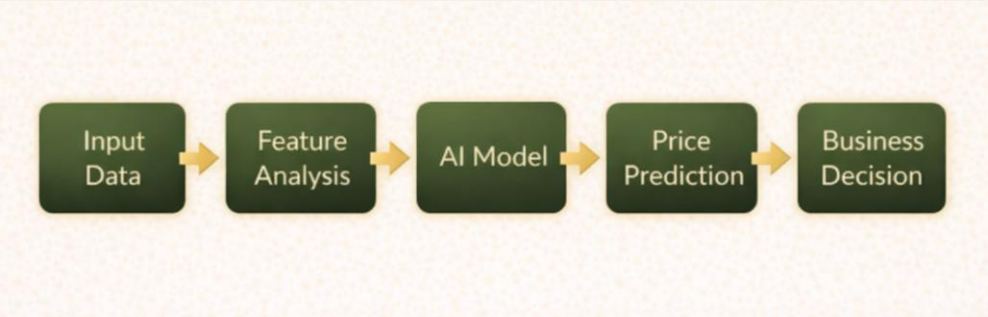
Achieving a better balance between maximizing revenue and improving the property's competitive positioning

Project Objectives

1. Analyze Airbnb data and extract patterns that influence pricing
2. Develop a machine learning model capable of predicting expected rental prices
3. Understand pricing differences across the ten cities under study
4. Measure the impact of spatial, structural, and service-related factors on rental value
5. Deliver a practical solution to improve pricing accuracy and maximize revenue

AI-Powered Pricing Solution

- AI-powered dynamic pricing for short-term rentals
- Accurate and dynamic price prediction
- Data-driven decision-making
- Improved efficiency and competitiveness



- +10–20% revenue improvement
- Faster pricing decisions
- Reduced human error

Study Scope & Data

Ten Cities Under Study

- New York
- Paris
- Rome
- Bangkok
- Rio de Janeiro
- Sydney
- Istanbul
- Hong Kong
- Cape Town
- Mexico City

Key Variables

Property Type: Entire apartment, hotel room, villa, and others

Bedrooms: Number of bedrooms and accommodation capacity

Guest Count: Maximum occupancy limit

Geographic Location: Latitude and longitude coordinates

Amenities: WiFi, kitchen, parking, and more

Data Source: Maven Analytics — Airbnb Listings & Reviews Dataset

Scientific & Technical Methodology

Data Understanding & Exploratory Analysis → Data Cleaning & Preprocessing → Feature Engineering & Indicator Derivation → Model Building & Training → Evaluation & Validation

Regression Model

A regression model was chosen because the target variable (price) is a continuous numerical variable. This allows understanding the relationship between explanatory features and expected rental value. The LGBMRegressor model was selected for its efficiency with large datasets.

Integrated Pipeline

An integrated Pipeline was built to ensure systematic handling of numerical and categorical data within a single reusable and extensible environment.

scikit-learn | LightGBM | StandardScaler | One-Hot Encoding

Project Workflow

DATA COLLECTION
Airbnb data from 10 global cities

↓

DATA PREPROCESSING
Data cleaning + Missing values + Outlier handling

↓

Feature Engineering
Location + Property + Amenities

Transformation & Encoding
Log1p + Scaling + Encoding

Machine Learning
Pipeline + Model Training

↓

Model Evaluation
MAE + RMSE + R^2

Deployment
Flask + Render

User Interface
Interactive Web Application

↓

Interactive Map
Spatial property visualization

Result Interpretation
Why this price?

AI Smart Assistant
AI Chatbot — OpenAI

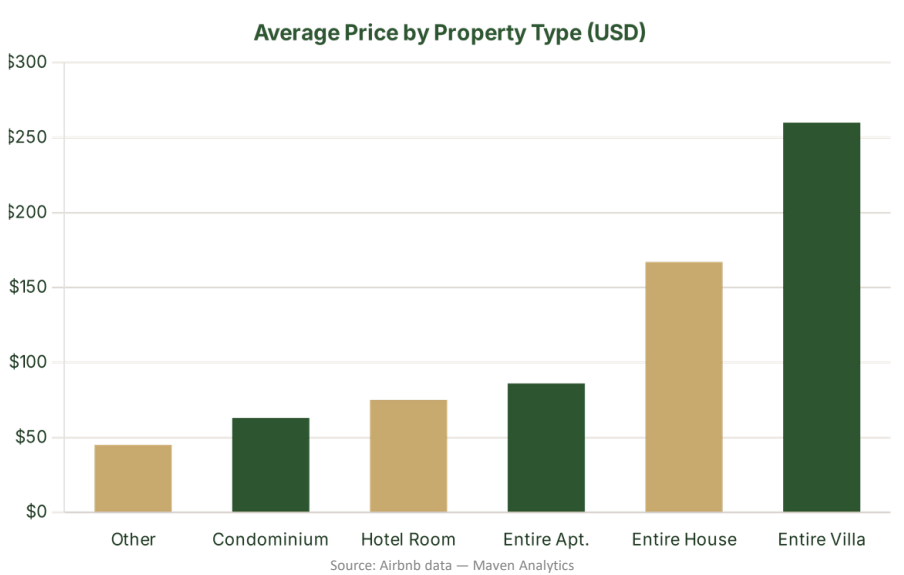
↓

Final System — AI-Powered Intelligent Price Prediction System

Feature Engineering: The Core Foundation

Factors Affecting Price

- **Distance from City Center:** Proximity to key areas increases rental value
- **Property Type:** Entire villa, apartment, hotel room, residential unit
- **Room Type:** Private room, shared room, entire unit
- **Bedrooms:** Number of bedrooms and guest capacity
- **Amenities:** WiFi, kitchen, parking, pool
- **City:** As a contextual variable reflecting local market dynamics



Price vs Distance Analysis

Proximity Premium

Properties closer to the city center consistently command higher rental prices across all ten cities.

Inverse Trend

A clear inverse relationship exists between distance and price, though significant variation reflects property-specific features.

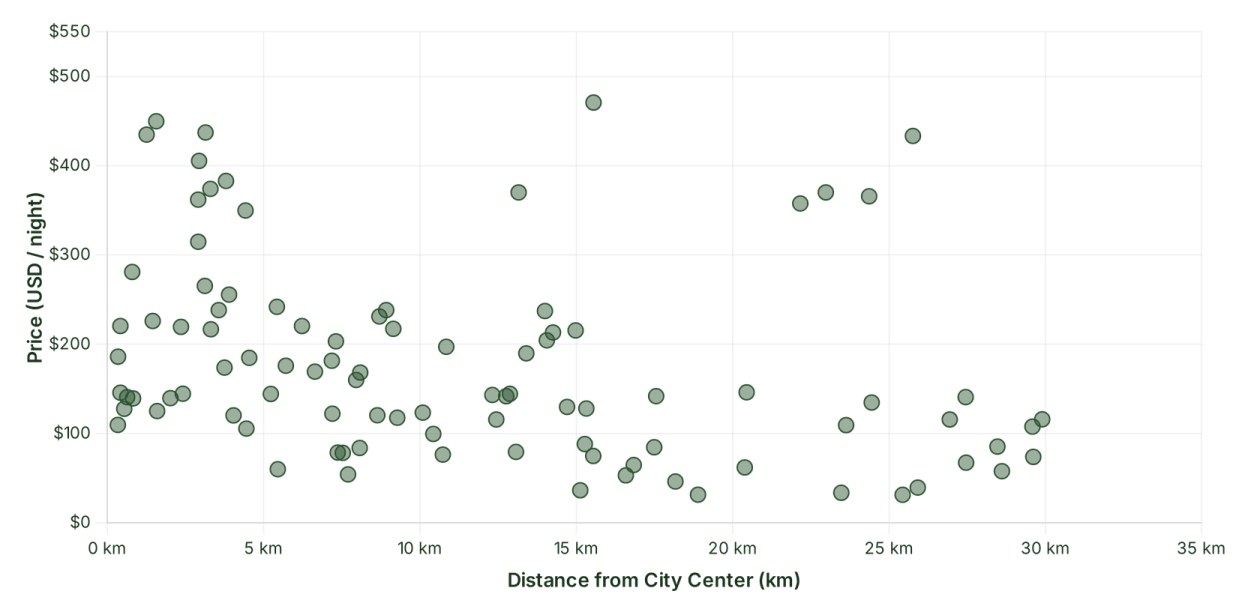
Premium Outliers

High-value outliers at greater distances indicate luxury villas and unique properties that defy the general trend.

GENERAL TREND

Inverse

Rental Price (USD) vs Distance from City Center (km)



Source: Airbnb data via Maven Analytics — 10 cities

Top 10 Most Expensive Cities

MOST EXPENSIVE

New York

Highest median rental price at $100/night

Lowest city price

Bangkok

Lowest median price at $32/night

PRICE GAP

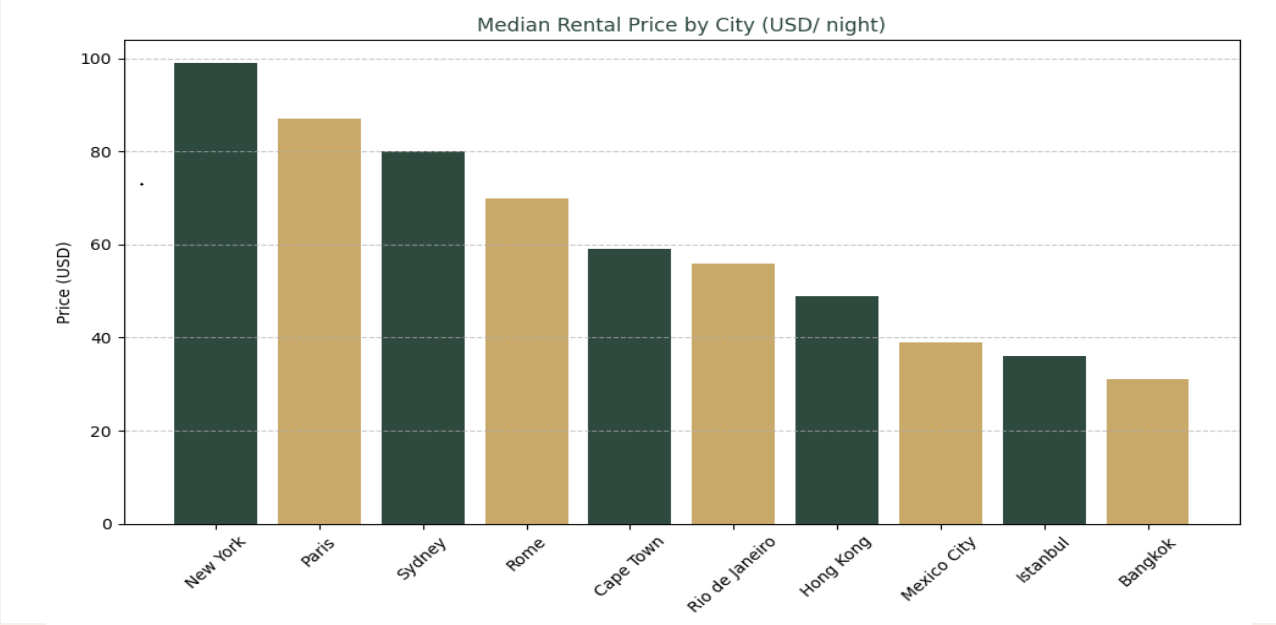
3.12x

Difference between highest and lowest city

KEY INSIGHT

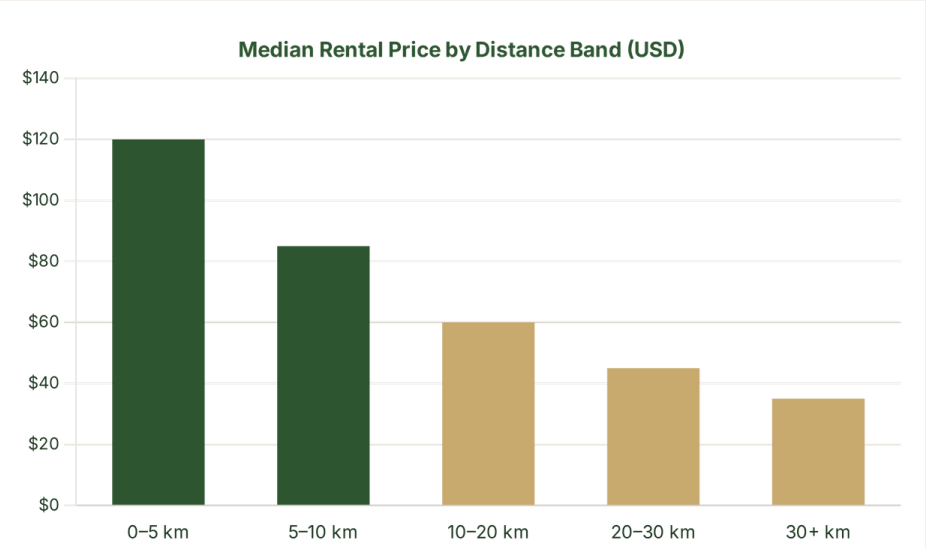
Market Diversity

City context is a critical pricing factor



Source: Airbnb data — Maven Analytics

Price Distribution by Distance Band



Source: Airbnb data — Maven Analytics

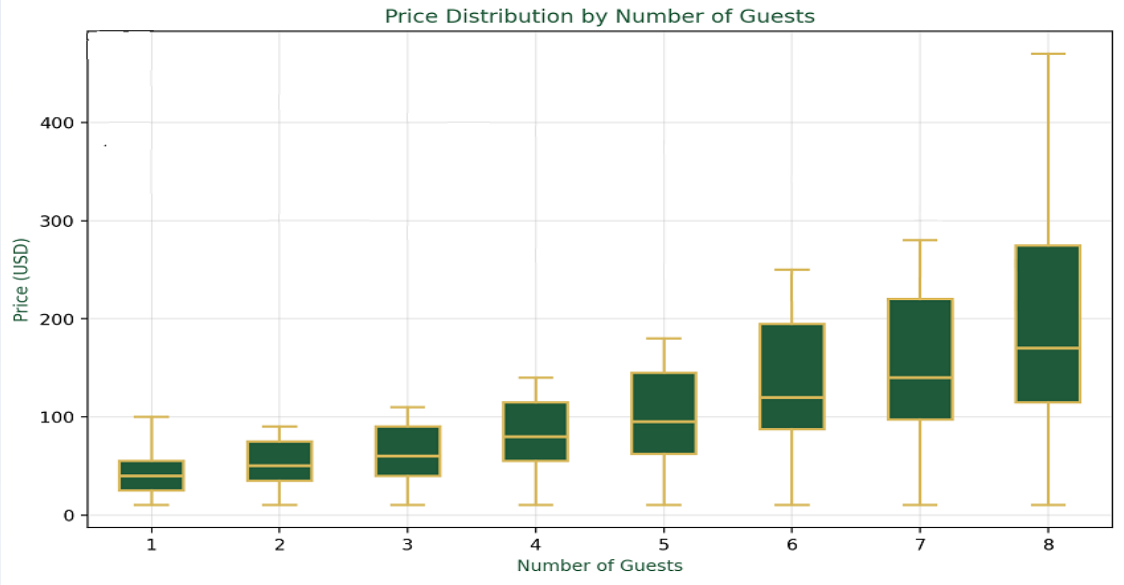
Median Price by Distance Category

Distance Band	Median Price	Relative
0 – 5 km	$120	Highest
5 – 10 km	$85	High
10 – 20 km	$60	Medium
20 – 30 km	$45	Low
30+ km	$35	Lowest

Properties within 5 km of the city center command a **3.4x premium** over those beyond 30 km, confirming location as the primary price driver.

Guest Capacity vs Price

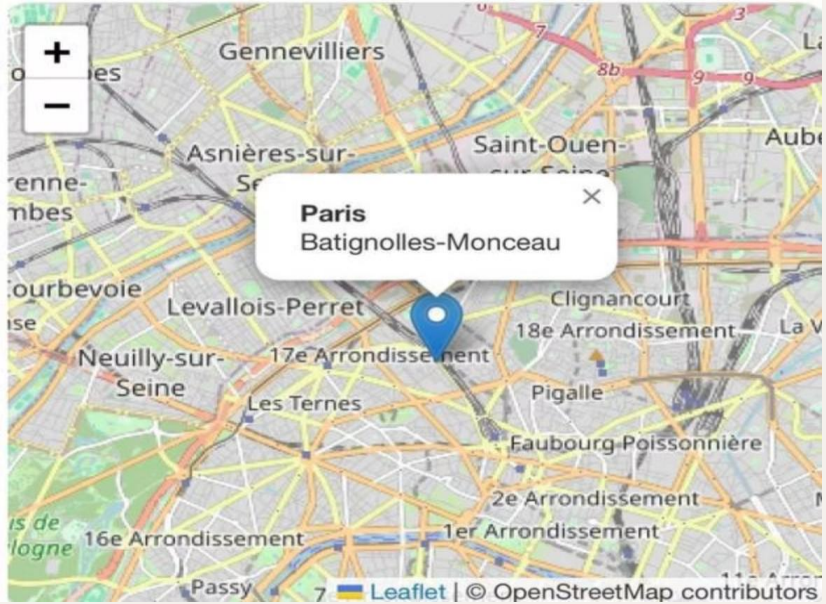
- A clear positive relationship exists between guest capacity and price.
- Median prices increase consistently with higher capacity.
- Price variability widens for larger listings.



💡 Key Insight

Higher capacity → higher prices with greater variability

Interactive Property Map



Map Features

Color-Coded Markers: Each city has a unique color for easy identification

Location base-Based Filtering: Automatically filters properties using latitude and longitude

Dynamic Map Navigation: Map content updates instantly based on user navigation

City Filter: Filter properties by city for focused analysis

Zoom & Pan: Navigate seamlessly across all ten cities worldwide

Technology: Built with Folium (Python) + Leaflet.js for interactive geospatial visualization

Model Evaluation Results

R^2 SCORE

0.86

The model explains 86% of the variance in rental prices

MAE

0.33

Mean Absolute Error on the log-transformed scale

RMSE

0.44

Root Mean Squared Error indicating prediction consistency

What This Means

An R^2 of 0.86 indicates strong predictive power. The model captures the vast majority of pricing patterns across all ten cities, making it a reliable tool for decision support.

Practical Impact

Low MAE and RMSE values on the log scale translate to predictions that are typically within 15–20% of the actual rental price — sufficient for operational pricing decisions.

AI Smart Assistant (Chatbot)

An AI-powered chatbot was integrated into the web application to provide natural language explanations of predicted prices and support decision-making through conversational interaction.

- **Natural Language:** Ask questions in plain language about pricing
- **Context-Aware:** Understands property features and market context
- **Actionable Insights:** Provides recommendations to optimize pricing
- **Multi-City Support:** Covers all ten cities in the dataset

Holiday Homes AI Assistant
Online

Why is the predicted price for my apartment in Paris $130/night?

Your apartment in Paris is priced at $130/night due to several factors: it's located 2.3 km from the city center (premium location), has 2 bedrooms accommodating 4 guests, and offers WiFi + kitchen amenities. Paris properties in this zone typically range $110–$160/night.

How can I increase the rental value?

Based on the model's feature importance analysis, adding amenities like parking or a balcony could increase your predicted price by 10–15%. Also, properties with "Superhost" status tend to command a 12% premium in Paris.

Powered by OpenAI API

SMART RENTAL DASHBOARD

Holiday Homes

Choose the city, neighbourhood, listing details, and amenities to estimate the nightly price and total stay cost.

Model is off

PROJECT ACCESS

Professional access to the pricing workspace

Sign in to open the full prediction dashboard, review listing details, and work inside the protected project area.

Secure entry

Password protection keeps the project gate simple and controlled.

Fast workflow

After login, you return directly to the pricing form and map tools.

Login

Enter the password to continue to the project dashboard.

Password

Enter your password

Log In

Render | Project Link

https://rent-price-prediction.onrender.com

Value Proposition & Business Impact

- Fast and accurate price estimation within seconds
- Simple and user-friendly interface
- Clear explanations based on real data
- Enables confident and informed decision-making

- Subscription-based SaaS for property owners
- Licensing solutions for real estate companies
- Paid API access for platform integration
- Enterprise-level customized solutions

- Enhances pricing accuracy compared to traditional methods
- Increases occupancy rate and revenue
- Reduces reliance on manual estimation
- Supports data-driven strategic decisions

- Optimizes pricing strategies
- Reduces vacancy periods
- Maximizes annual revenue
- Improves overall investment performance

Conclusion & Future Development

Key Findings

- **High Accuracy:** The model achieved $R^2 = 0.86$, explaining 86% of rental price variance across ten cities
- **Location is Key:** Distance from city center proved to be the most influential pricing factor
- **Multi-City Coverage:** A single model successfully handles diverse markets from Hong Kong to Mexico City
- **Practical Framework:** Complete pipeline from data to deployment, ready for operational use

Future Development

1. **Real-Time Data Integration:** Live market data for accurate and up-
2. **Dynamic Pricing Engine:** Automatically adjusts prices based on demand and trer
3. **Mobile Application:** Native iOS/Android app for on-the-go pricing decisions
4. **Multi-Language Support:** Supports multiple languages for global accessibility
5. **Market Expansion:** Expand to new markets and property types

Special Thanks

TO OUR TRAINERS

We appreciate the guidance, support, and, collaboration that made this project possible.

Lead AI Trainer
Ola El-Sheikh

Training Facilitator
Ahmad Deyab

Technical Support
Hisham El-Sisi

Technical Language Instructor
Omar Kamel

AI Tools Instructor
Nour Galal

Powered by

mystro

THANK YOU!
Q & A (FOR ANY QUESTIONS)